	loop.
2(a)	Gravitational potential at infinity is taken to be zero. Due to the attractive nature of the gravitational force, work done by an external agent to bring any mass from infinity to that point is always negative. Hence the potential at any point must always be negative.
2(b)(i)	Gravitational potential at the surface of the planet $\phi_{\text{surface}} = -\frac{GM}{R_{\text{planet}}}$ $= -\frac{6.67 \times 10^{-11} \times 6.2 \times 10^{23}}{3.4 \times 10^{3} \times 10^{3}}$ $= -1.216 \times 10^{7}$ $\approx -1.22 \times 10^{7} \text{ J kg}^{-1}$

No.	Solution
2(b)(ii)	The minimum kinetic energy required of the rock is equivalent to the gain in gravitational potential energy by the rock in leaving planet to escape to infinity. Loss in KE = Gain in GPE $\text{KE}_{\text{esc}} + \text{GPE}_{\text{surface}} = \text{KE}_{\infty} + \text{GPE}_{\infty}$ $\frac{1}{2}mv_{\text{esc}}^{2} + m\phi_{\text{surface}} = 0$ $\frac{1}{2}v_{\text{esc}}^{2} - 1.216\times10^{7} = 0$ $\therefore v_{\text{esc}} = \sqrt{2\left(1.216\times10^{7}\right)}$ $= 4.93\times10^{3}\ \text{m s}^{-1}$ Since the speed of the rock is $3.8\times10^{3}\ \text{m s}^{-1}$ which is less than the escape speed needed to leave the planet, the rock will return to the surface of the planet.
3(a)	The internal energy U of an ideal gas is the sum of a random distribution of kinetic